

BCA-5003: COMPUTER NETWORK

UNIT V — COMPLETE DETAILED NOTES + TOPIC-WISE PYQ ANSWER BOOK

Transport Layer • Session, Presentation & Application Layers • Routing • Cryptography • Email Protocols

Unit-I format: Index → detailed concepts → diagrams/tables → topic-wise PYQs with years → long model answers → revision plan

Prepared from the supplied Unit-V material. The notes use corrected technical terms for safe, high-mark examination answers.

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High-score strategy

1. For every OSI-layer answer, state layer number, main purpose, functions, common protocols, and one practical example.
2. For TCP connection questions, draw the three-way handshake and four-segment close procedure.
3. For cryptography, use plaintext → encryption → ciphertext → decryption → plaintext diagrams.
4. For routing, distinguish routing from forwarding, then explain static/adaptive algorithms and table fields.
5. For email/MIME, write protocol role and architecture diagrams, not only full forms.

Connection established; data may now flow in both directions.

1. Client performs an active open and sends SYN with initial sequence number x .
2. Server responds with SYN+ACK: it acknowledges $x+1$ and announces its own initial sequence number y .
3. Client sends ACK for $y+1$. Both sides now know the other side is reachable and sequence spaces are synchronized.

Connection termination

TCP close is normally a four-segment exchange because each direction closes independently.

```
Client Server
FIN, Seq = u ████████████████████████████████████████████████████████████
██████████████████████████████████████████████████████████████████████████ ACK, Ack = u + 1
██████████████████████████████████████████████████████████████████████████ FIN, Seq = v
ACK, Ack = v + 1 ████████████████████████████████████████████████████████████
```

Connection closed after required state/timer handling.

A reset (RST) can abort a connection immediately in exceptional/error cases; it is not the normal orderly closing procedure.

4. Session and Presentation Layers

4.1 Session Layer — Layer 5

The Session Layer establishes, manages, synchronizes, and terminates dialogs/sessions between application processes. In modern Internet implementations, many session functions are handled by application frameworks and protocols rather than a separate universal Session-Layer protocol, but the OSI functions remain important for examination answers.

Function	Explanation
Session establishment	Creates a logical dialogue/session and agrees parameters before extended communication.
Session maintenance	Maintains/coordinates the active dialogue during communication.
Session termination	Closes the dialogue and releases associated resources.
Dialog control	Manages which side may transmit and supports half-duplex/full-duplex dialog conventions.
Synchronization	Adds checkpoints in a long operation, enabling restart from a checkpoint after interruption.
Recovery support	Helps coordinate resumption after failures using synchronization points.

Checkpoint diagram

Long transfer: Start ■■ data ■■ [Checkpoint 1] ■■ data ■■ [Checkpoint 2] ■■ failure
Recovery: restart from Checkpoint 2 instead of restarting from Start.

4.2 Presentation Layer — Layer 6

The Presentation Layer is concerned with data representation, syntax, and transformation. It ensures that systems with different internal data formats can exchange information correctly.

Function	Explanation
Translation	Converts data formats/character sets between systems, such as ASCII/Unicode and different numeric representations.
Compression	Reduces data size to save bandwidth/storage; receiver decompresses it.
Encryption/decryption	Transforms readable plaintext into ciphertext and restores it for confidentiality.
Serialization/format representation	Encodes structured data in transferable formats; examples include common image/document/data formats.

Presentation-layer transformation diagram

Sender data format → translate/compress/encrypt → network transfer → decrypt/decompress/translate → receiver data format

5. Application Layer and Services

The Application Layer is Layer 7 and provides network services directly to end-user applications. It is not the user application itself; it is the layer/interface through which applications use network services. Common Application-Layer protocols include HTTP/HTTPS, SMTP, DNS, FTP, SSH, DHCP, SNMP, and others.

Service	Explanation and examples
Web service	HTTP/HTTPS allow browser-client and web-server communication for pages, APIs, media, and web applications.
Email service	SMTP sends/transfers mail; IMAP or POP3 retrieves mail from server; MIME allows non-ASCII and attachment content.
File transfer	FTP/SFTP/SCP and related protocols transfer/manage files between systems.
Remote access	SSH provides secure remote command access; Telnet is an older insecure remote terminal protocol.
Name resolution	DNS maps names such as example.com to IP addresses and provides other domain records.
Directory service	LDAP and related systems manage/query directory information: users, groups, devices, permissions.
Network management	SNMP allows monitoring/management of network devices.
Messaging/collaboration	Protocols/services support chat, presence, conferencing, and real-time collaboration.
Remote procedure calls	RPC lets a program invoke a function/service on another machine as part of distributed computing.

Application-service diagram

[Browser] ■ HTTP(S) ■■ [Web server]
 [Mail client] ■ SMTP/IMAP/POP3 ■■ [Mail server]
 [User/device] ■ DNS query ■■ [DNS resolver/server]
 [Admin terminal] ■ SSH ■■ [Remote server/router]

6. Email and Common Application Protocols

SMTP

SMTP (Simple Mail Transfer Protocol) is used to submit and relay email from a sender's mail client/server to the recipient's mail server. It is a push/transfer protocol; it is not normally used by an end user to retrieve stored mail. For retrieval, POP3 or IMAP is used.

Email architecture

Sender MUA → MSA → MTA ■■■ SMTP ■■■■ recipient MTA → MDA → mailbox → IMAP/POP3 → recipient MUA
 MUA = Mail User Agent; MSA = Mail Submission Agent; MTA = Mail Transfer Agent; MDA = Mail Delivery Agent.

MIME

MIME stands for Multipurpose Internet Mail Extensions. It extends the original email format so messages can carry non-ASCII text, attachments, images, audio, video, documents, and multiple message parts. MIME does not itself deliver email; it labels/encodes content for email transport.

Important MIME headers

Header	Purpose / example
MIME-Version	Indicates MIME formatting, e.g. MIME-Version: 1.0.
Content-Type	Describes media type/subtype, e.g. text/plain, text/html, image/jpeg, application/pdf, multipart/mixed.
Content-Transfer-Encoding	Specifies safe representation for transport, e.g. base64 or quoted-printable.
Content-Disposition	Describes presentation, e.g. attachment; filename="notes.pdf".

Other important protocols

Protocol	Main role	Typical port note
HTTP / HTTPS	Web requests/responses. HTTPS is HTTP protected by TLS.	80 / 443
FTP	File transfer; classic FTP uses separate control and data connections.	21 control; 20 active-mode data historically
SSH	Secure remote login/command execution and tunnelling.	22
Telnet	Older remote terminal protocol; sends data including credentials without modern encryption	23
DNS	Name-to-address and other domain record lookup.	53 UDP/TCP

Protocol	Main role	Typical port note
POP3	Downloads/retrieves mail from server, often for local storage.	110; 995 with implicit TLS
IMAP	Synchronizes/accesses mail while keeping mailboxes on server.	143; 993 with implicit TLS

7. Cryptography, Encryption and Decryption

Cryptography is the science and practice of protecting information through mathematical algorithms and keys. It protects data stored on systems and data transmitted over networks. Cryptography supports confidentiality, integrity, authentication, and non-repudiation when appropriate mechanisms are used.

Security goals

Goal	Meaning
Confidentiality	Only authorized parties can read information. Encryption supports this goal.
Integrity	Unauthorized alteration is detected. Hashes/MACs/digital signatures support this.
Authentication	Verifies identity or origin of a user/message/system.
Non-repudiation	A sender cannot credibly deny a signed action/message, subject to proper key control and legal context.

Symmetric-key encryption

Symmetric encryption uses the same shared secret key for encryption and decryption. It is fast and suitable for large amounts of data. Its main challenge is secure key distribution: both parties must obtain and protect the same secret key. AES is a common modern symmetric algorithm; DES is historically important but not considered secure for modern use.

Plaintext ■■ Encrypt with shared secret key K ■■■ Ciphertext ■■ Decrypt with same K ■■■ Plaintext

Asymmetric/public-key encryption

Asymmetric cryptography uses a mathematically related key pair: public key and private key. For confidentiality, a sender encrypts using recipient's public key and only recipient's private key can decrypt. Asymmetric methods are slower than symmetric encryption but solve key-distribution and digital-signature problems. RSA and elliptic-curve cryptography are common examples.

Plaintext ■■ Encrypt with receiver public key ■■■ Ciphertext ■■ Decrypt with receiver private key ■■■ Plaintext

Hash functions

A cryptographic hash function converts input of arbitrary length into a fixed-length digest. Hashing is one-way: it is not encryption and is not reversed by "decryption." Hashes help verify integrity and are used with salts/slow password-hashing schemes for password storage. Examples include SHA-256; weak legacy hashes should not be used for security.

Digital signatures

Digital signatures normally use asymmetric cryptography. Sender hashes the message and signs the hash using the sender's private key. Anyone can verify the signature using the sender's public key. This supports authenticity, integrity, and non-repudiation properties.

Cryptosystem components

Component	Meaning
Plaintext	Original readable information to be protected.
Encryption algorithm	Mathematical process that converts plaintext to ciphertext using key material.
Encryption key	Key used as input to encryption operation.
Ciphertext	Unreadable/protected output produced by encryption.
Decryption algorithm	Process that converts ciphertext back to plaintext using the correct key.
Decryption key	Key used to recover plaintext; same as encryption key in symmetric systems, different/private key in public-key confidentiality.
Key space	Set of all possible keys for a cryptosystem.

8. Routing Algorithms and Routing Tables

Routing is the process of selecting a path for packets across interconnected networks. Forwarding is the router's local action of sending a received packet to the appropriate next-hop interface according to its routing/forwarding table. The "best" route generally means lowest cost according to a metric, not always simply the fewest hops.

Routing process diagram

Source host → Router A → Router B → Router C → Destination network
routing algorithm builds routes; forwarding table chooses next hop for each packet

Routing algorithm categories

Type	Description	Examples/notes
Static / non-adaptive	Routes are manually configured or fixed and do not automatically react to topology changes.	Static routes, fixed routing. Suitable for small, stable networks.
Dynamic / adaptive	Routers exchange/update routing information and react to failures/topology changes.	Distance vector, link state, path vector.
Flooding	Forwards packet on all outgoing links except incoming link; uses controls such as hop count to prevent loops.	Useful for discovery, but inefficient for data delivery.

Major dynamic approaches

Approach	Main idea	Typical protocol
Distance vector	Router shares distance/cost to destinations with neighbours; chooses best neighbour route.	RIP uses hop count metric.
Link state	Router learns network topology and link costs, then computes shortest paths.	OSPF.
Path vector	Routes carry path/Autonomous System information; policy based inter-domain selection.	BGP.

Routing table

A routing table is a data structure maintained by a router/host. It contains routes that tell the device how to reach destination networks. A route typically specifies destination prefix, subnet mask/prefix length, next hop, outgoing interface, metric, and source/type of route.

Example routing table

Destination prefix	Next hop	Interface	Metric	Meaning
192.168.1.0/24	Directly connected	LAN1	0	Local LAN.
10.10.0.0/16	192.168.1.254	LAN1	10	Send to next-hop router.
0.0.0.0/0	192.168.1.1	LAN1	100	Default route for unknown destinations.

9. Hamming Distance and Hamming Code

Hamming distance is the number of bit positions at which two equal-length binary strings differ. Example: distance between `1011011` and `1001110` is 3 because three positions differ. The minimum Hamming distance of a code determines its error-detection/correction capability.

Useful rule

A code with minimum distance $d_{\min}$ can detect up to $d_{\min} - 1$ errors and can correct up to $\lfloor (d_{\min} - 1)/2 \rfloor$ errors. Basic Hamming codes have minimum distance 3 and can correct one-bit errors.

Hamming (7,4) code layout

Position: 1 2 3 4 5 6 7
Role: P1 P2 D1 P4 D2 D3 D4

Even parity groups:
P1 checks positions 1, 3, 5, 7
P2 checks positions 2, 3, 6, 7
P4 checks positions 4, 5, 6, 7

Received word example: 1011011 (even parity, positions numbered left to right)

Positions: 1 2 3 4 5 6 7
Bits: 1 0 1 1 0 1 1

P1 group (1,3,5,7) = 1,1,0,1 → odd → fails
P2 group (2,3,6,7) = 0,1,1,1 → odd → fails
P4 group (4,5,6,7) = 1,0,1,1 → odd → fails
Syndrome $P_4P_2P_1 = 111$ = 7, so bit 7 is wrong.
Corrected codeword: 1011010.

10. Topic-Wise PYQ Answer Bank

The following model answers correspond to questions visible in the Unit-V source. They are organized after the detailed notes, exactly as requested.

Q1. Discuss the services offered by the Application Layer.

Asked: 2023–24, 2024–25, 2025–26

Answer

The Application Layer is the seventh and highest layer of the OSI model. It provides network services directly to end-user applications. It acts as the interface through which application software uses the underlying network.

- 1. File transfer and management:** Protocols such as FTP/SFTP enable upload, download, remote file access, directory listing, and file management.
- 2. Email service:** SMTP transfers email between clients/servers; POP3 or IMAP retrieves messages for users. MIME enables attachments and multimedia mail content.
- 3. Web service:** HTTP/HTTPS support communication between browsers and web servers for websites, web applications, APIs, images, and multimedia resources.
- 4. Remote access:** SSH enables secure terminal access to remote systems. Telnet is an older remote terminal protocol but is insecure because it does not provide modern encryption.
- 5. Name service:** DNS translates human-readable domain names into IP addresses, enabling users to access Internet services by names rather than numeric addresses.
- 6. Directory service:** LDAP-type services store and provide controlled access to users, groups, devices, and organisational directory information.
- 7. Network management:** SNMP and related tools support monitoring and management of network devices.
- 8. Messaging, collaboration and RPC:** Application services support chat, presence, conferencing, remote procedure calls, and distributed applications.

Conclusion: The Application Layer makes network resources useful to users by delivering services such as web access, email, file transfer, remote login, name resolution, and collaboration.

Q2. Explain cryptography used for network security and privacy.

Asked: 2023–24, 2025–26

Answer

Cryptography is the use of mathematical algorithms and keys to protect information and communication from unauthorized access or alteration. It is used in secure web browsing, online payments, VPNs, digital signatures, authentication systems, and protected data storage.

Cryptography provides **confidentiality** so only authorised parties can read information; **integrity** so unauthorised changes are detected; **authentication** so identities/origin can be verified; and **non-repudiation** so a validly signed sender cannot easily deny an action/message.

Symmetric-key cryptography: the same secret key encrypts and decrypts data. It is fast and appropriate for bulk data encryption. AES is a standard modern example. Its main issue is distributing the shared key securely.

Asymmetric-key cryptography: uses a public/private key pair. For confidentiality, sender encrypts using receiver's public key and receiver decrypts using private key. RSA and ECC are common examples. It is useful for key exchange and digital signatures but usually slower than symmetric encryption.

Hash functions: create fixed-length one-way digests for integrity checking. Hashing is not encryption and cannot normally be reversed. Hashes are used in integrity verification, signatures, and password-storage systems with secure salting/slow hashing.

Digital signature: sender signs a message hash with private key; verifier checks using public key. This supports authenticity and integrity.

Conclusion: Modern secure systems commonly combine asymmetric cryptography for identity/key establishment, symmetric encryption for bulk data, and hashes/signatures for integrity and authentication.

Q3. Explain encryption and decryption methods.

Asked: 2024–25

Answer

Encryption is the process of converting readable data, called plaintext, into unreadable protected data, called ciphertext, using an algorithm and key. Decryption is the reverse process: it uses the correct key and algorithm to restore ciphertext to plaintext for an authorised recipient.

Symmetric method: one shared secret key is used for both encryption and decryption. Diagram: Plaintext → Encrypt using key K → Ciphertext → Decrypt using key K → Plaintext. AES is a common example. It is fast but needs secure key sharing.

Asymmetric method: uses a related public/private key pair. For confidentiality: Plaintext → Encrypt with receiver public key → Ciphertext → Decrypt with receiver private key → Plaintext. RSA/ECC are examples. This solves public key distribution but is slower for large data.

Hybrid method: Practical systems such as HTTPS normally use public-key methods to authenticate/exchange a temporary symmetric session key, then symmetric encryption for actual bulk data because it is faster.

Conclusion: Encryption provides confidentiality during storage/transmission; decryption permits only the correct key holder to read data.

Q4. Explain routing algorithms and write short notes on routing table.

Asked: 2024–25

Answer

Routing is path selection for packets across interconnected networks. A routing algorithm determines the best route according to metric/cost such as hop count, delay, bandwidth, reliability, or administrative policy. Forwarding then sends a packet to the selected next hop.

Static/non-adaptive routing: routes are manually configured and do not automatically change with failures or traffic. It is simple and suitable for small stable networks but lacks adaptability.

Dynamic/adaptive routing: routers update routes based on network information. Distance-vector algorithms exchange distance/cost information with neighbours; RIP is a classic example. Link-state algorithms build topology knowledge and compute shortest paths; OSPF is an example. Path-vector routing is used between autonomous systems; BGP is the key example.

Routing table: a router's data structure containing destination network prefixes, prefix lengths/masks, next-hop addresses, outgoing interfaces, metrics, and route source/type. The router uses longest-prefix matching to select the most specific suitable route. A default route is used when no more specific entry matches.

Conclusion: Routing algorithms create/maintain route knowledge; routing tables are used for actual packet-forwarding decisions.

Q5. Explain duties/functions of the Transport Layer.

Asked: 2016, 2017, 2022–23

Answer

The Transport Layer is Layer 4 of the OSI model and provides process-to-process delivery. It ensures that data reaches the correct application process on the destination host, not only the correct machine.

1. Port addressing: source/destination ports identify applications and enable multiple applications to share one host/network connection.

2. Segmentation and reassembly: large application messages are divided into segments and correctly reassembled at destination.

3. Connection control: may establish/manage/release a connection-oriented service or send independent connectionless datagrams.

4. Reliability: TCP uses sequence numbers, acknowledgements, checksums, timers and retransmission to recover from losses/corruption and deliver bytes in order.

5. Flow control: prevents a fast sender from overwhelming a slower receiver using buffering/window mechanisms.

6. Congestion control: TCP adjusts sending rate based on network congestion to reduce loss and overload.

7. Multiplexing/demultiplexing: combines data from many source processes and sends received data to appropriate destination process.

Conclusion: The Transport Layer makes end-to-end application communication reliable, efficient, and properly addressed.

Q6. Explain the components of cryptography/cryptosystem.

Asked: 2022–23

Answer

A cryptosystem is a complete set of components used to protect information through encryption and decryption.

Plaintext: original readable information to be protected.

Encryption algorithm: mathematical procedure that takes plaintext and an encryption key and produces ciphertext.

Encryption key: secret/private/public key material used as input to encryption.

Ciphertext: scrambled output that should be unreadable without the appropriate decryption key.

Decryption algorithm: mathematical procedure that takes ciphertext and correct decryption key and restores plaintext.

Decryption key: key used to recover the plaintext. In symmetric cryptography it is the same shared secret key; in asymmetric confidentiality it is usually the recipient's private key.

Key space: set of all possible valid keys for a cryptographic scheme. Large key spaces make brute-force attacks harder.

Attacker/interceptor: unauthorised entity attempting to recover plaintext or keys. Secure cryptography assumes algorithms may be publicly known; security should depend on protecting keys, not hiding the algorithm.

Q7. Explain Hamming distance. Received 7-bit Hamming codeword is 1011011 using even parity. Is it correct? If not, locate error.

Asked: 2022–23

Answer

Hamming distance is the number of positions in which two equal-length binary strings differ. It is used to measure code separation and determine error-detection/correction ability. Basic Hamming code adds parity bits at positions 1, 2, 4, and so on to locate a single-bit error.

Assume the seven positions are numbered left to right. Received word: positions 1–7 = 1 0 1 1 0 1 1. With even parity, P1 checks 1,3,5,7 = 1,1,0,1, which has three 1s and fails. P2 checks 2,3,6,7 = 0,1,1,1, which has three 1s and fails. P4 checks 4,5,6,7 = 1,0,1,1, which has three 1s and fails.

Syndrome = $P4P2P1 = 111 = 7$. Therefore, bit position 7 is in error. Flip the seventh bit from 1 to 0. Corrected codeword = **1011010**. Hence received codeword is wrong and has a single-bit error at position 7.

Always state the numbering convention in an exam because some texts number from the right; the method is unchanged.

Q8. What is MIME? Explain standard content types of MIME.

Asked: 2019

Answer

MIME stands for Multipurpose Internet Mail Extensions. It extends the original Internet email format so email can carry non-ASCII text, attachments, images, audio, video, documents, and multiple content parts. MIME works with email systems; it does not replace SMTP for mail transport.

MIME uses headers such as MIME-Version, Content-Type, Content-Transfer-Encoding, and Content-Disposition. Content-Type contains a major type and subtype.

text: text/plain, text/html. Used for ordinary/plain or HTML text messages.

image: image/jpeg, image/png, image/gif. Used for image attachments/content.

audio: audio/mpeg, audio/wav. Used for sound content.

video: video/mp4, video/mpeg. Used for video content.

application: application/pdf, application/zip, application/octet-stream. Used for documents, binary files, executable-like generic data, etc.

multipart: multipart/mixed for text plus attachments; multipart/alternative for same content in formats such as text/plain and text/html; multipart/related for linked resources; multipart/form-data for form submissions.

message: message/rfc822 for an attached/embedded email message.

Conclusion: MIME labels and encodes content so mail clients know how to safely transfer and display different kinds of email data.

Q9. What is SMTP? Explain its components with example.

Asked: 2019

Answer

SMTP, Simple Mail Transfer Protocol, is the standard Application-Layer protocol for submitting and relaying email. It transfers outgoing messages from a user's mail client to a submission server and between mail servers until the recipient domain's server accepts the message. It uses TCP for reliable transfer.

Components: MUA (Mail User Agent) is the user's email client such as a webmail interface or mail app. MSA (Mail Submission Agent) receives submitted outgoing mail. MTA (Mail Transfer Agent) relays mail between servers using SMTP and DNS MX records. MDA (Mail Delivery Agent) places accepted mail in the recipient mailbox. The recipient then reads mail using IMAP or POP3 through a MUA.

Example flow: Alice composes a message in MUA → submits it to her MSA → MTA queries DNS for the recipient domain's mail exchanger → sends SMTP message to recipient MTA → MDA stores it in recipient mailbox → Bob opens MUA and retrieves/synchronizes mail using IMAP or POP3.

Conclusion: SMTP handles sending and relaying; IMAP/POP3 mainly handle user retrieval of received mail.

Q10. Write five Application-Layer protocols and explain each briefly.

Asked: 2018

Answer

SMTP: sends and relays electronic mail between mail clients/servers and mail servers.

HTTP/HTTPS: transfers web pages, APIs and web resources between browser/client and web server; HTTPS adds TLS protection.

FTP: transfers files between hosts. Classic FTP uses a control connection and a separate data connection.

DNS: resolves domain names to IP addresses and provides other DNS records needed for Internet services.

Telnet/SSH: provide remote terminal access. Telnet is legacy and unencrypted; SSH is the secure modern alternative.

Other acceptable examples include POP3, IMAP, DHCP, SNMP, NTP, LDAP, and TFTP. In answers, always write the full form plus one precise function.

Q11. Discuss difference between Presentation and Application Layer.

Asked in supplied Unit-V material

Answer

Presentation Layer: Layer 6. It handles data representation: translation between formats/character sets, compression/decompression, encryption/decryption, and data syntax. It normally does not directly offer web/email/file services to the user.

Application Layer: Layer 7. It provides network services directly to user applications, such as email, file transfer, web access, DNS, remote login, and network management.

In short: Presentation Layer makes data understandable and secure across unlike systems; Application Layer makes network services available to applications/users.

Q12. Write functions of Session Layer.

Asked in supplied Unit-V material

Answer

The Session Layer is Layer 5 of the OSI model. It establishes, manages, synchronizes, and terminates logical communication sessions between applications.

Session establishment: creates a session and agrees communication parameters before an extended dialogue.

Session maintenance: manages the active session and coordinates the dialogue while communication continues.

Dialog control: manages who may send and when; supports half-duplex/full-duplex conversational coordination.

Synchronization: inserts checkpoints in a long data transfer or transaction. If failure occurs, processing can resume from the latest checkpoint rather than beginning again.

Session termination: closes the session cleanly and releases resources.

Conclusion: These functions organize and recover long-running application conversations.

11. Final Revision Sheet

Must-practise diagrams

- Transport Layer segmentation and port-based delivery.
- TCP three-way handshake and TCP four-step termination.
- Session checkpoint/recovery diagram.
- Presentation transformation: translate/compress/encrypt → reverse at receiver.
- Application service diagram: browser/web server, mail client/mail server, DNS query, SSH remote access.
- SMTP mail-flow architecture: MUA → MSA → MTA → MDA → mailbox → IMAP/POP3 → MUA.
- Symmetric encryption, asymmetric encryption, hash, and digital-signature diagrams.
- Routing process and a sample routing table.
- Hamming (7,4) parity-position/checking layout.

Fast comparison tables

Concept	Remember this
TCP vs UDP	TCP is reliable, ordered, connection-oriented; UDP is connectionless, lightweight, best effort.
Session vs Presentation	Session manages dialogue/checkpoints; Presentation handles syntax, translation, compression, encryption.
Presentation vs Application	Presentation transforms data; Application provides services to applications/users.
Symmetric vs Asymmetric	Same shared key vs public/private key pair. Symmetric is faster; asymmetric helps with key exchange/signatures.
Encryption vs Hashing	Encryption is reversible with a key; cryptographic hashing is one-way and used for integrity.
Static vs Dynamic routing	Manual/fixed routes vs routes that update based on network information.
SMTP vs POP3/IMAP	SMTP sends/relays email; POP3/IMAP retrieves/accesses mailbox email.

7-Day Unit-V 10-CGPA Study Plan

Day	Focus	Output
1	Transport layer	Write functions and TCP vs UDP table from memory.
2	TCP handshake/release	Draw handshake and close diagrams; practise Q5.
3	Session + Presentation	Prepare Q11/Q12 and all layer functions.
4	Application services + protocols	Write Q1/Q10; learn protocol roles and ports.
5	SMTP + MIME	Draw email architecture; learn MIME header/content types.
6	Cryptography + routing	Write Q2/Q3/Q4/Q6; revise diagrams.
7	Hamming + complete mock	Solve Hamming numerical; write all important PYQs under time limit.

Accuracy reminders

- The Application Layer is not the same thing as a user application; it provides services to applications.
- SMTP primarily sends/relays mail. POP3/IMAP are used to retrieve/access mailboxes.
- Hashing is not encryption. A secure hash is designed to be one-way.
- Encryption supports confidentiality; integrity/authentication/non-repudiation need appropriate hashes, MACs, signatures, and key management.
- Routing chooses paths; forwarding sends packets using a route/forwarding table.
- A “best route” is lowest metric/policy cost, not necessarily fewest hops.
- TCP has orderly close using FIN/ACK; RST is normally an abrupt reset.

Final high-score rule: learn definitions, reproduce the diagrams without looking, write the year-wise PYQs in your notebook, and revise the formula/terms tables before the exam.